

queue

FIFO

First In First Out

VOV



TO DO LIST

1.

2.

3.

4.

5.

enqueue

dequeue

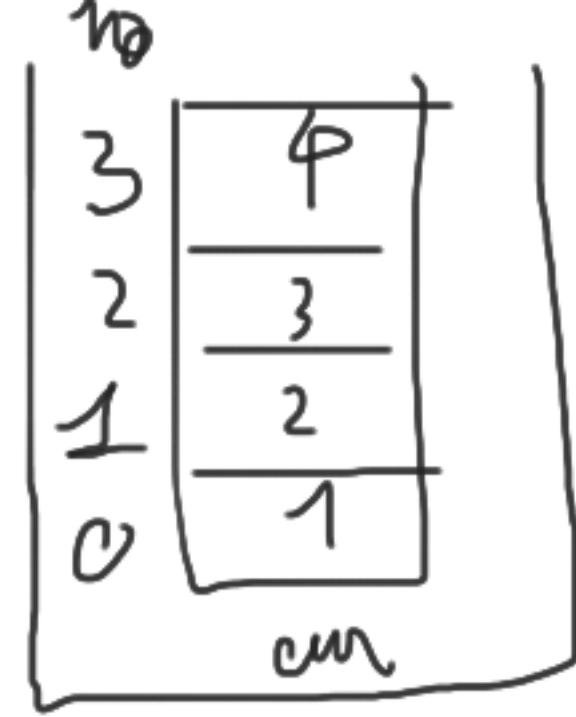
stacks

LIFO

Last In First Out

push

pop



arr[0]

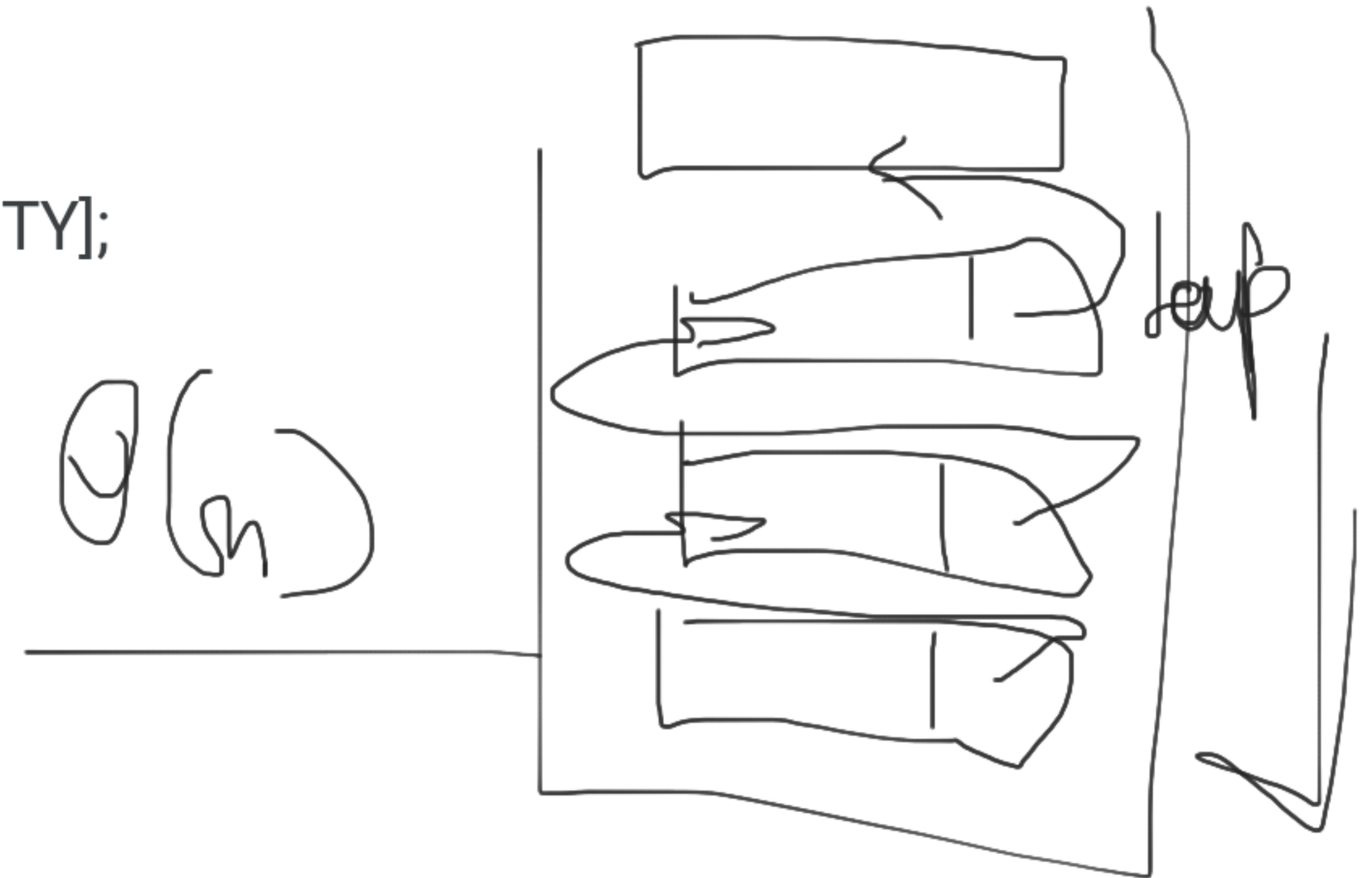
arr[10]

const int CAPACITY = 100;

```
class Stack {
    Person[ ] person = new Person[CAPACITY];
    int size;
}
```

// SỐ NGƯỜI STACK LƯU ĐƯỢC
 // SIZE: KÍCH CỠ THỰC SỰ STACK

LINKED LIST



push()

Add new item into to the stack. If the stack is full => Overflow condition

Algorithm:

begin

if stack if full

return

endif

else

increment top

stack[top] assign value

end else

end procedure

pop()

Remove an item from the stack. The items are popped in the top, if the stack is empty, then it is said to be an Underflow condition

Algorithm:

begin

if stack is empty

return

endif

else

store value of stack[top]

decrement top

return top

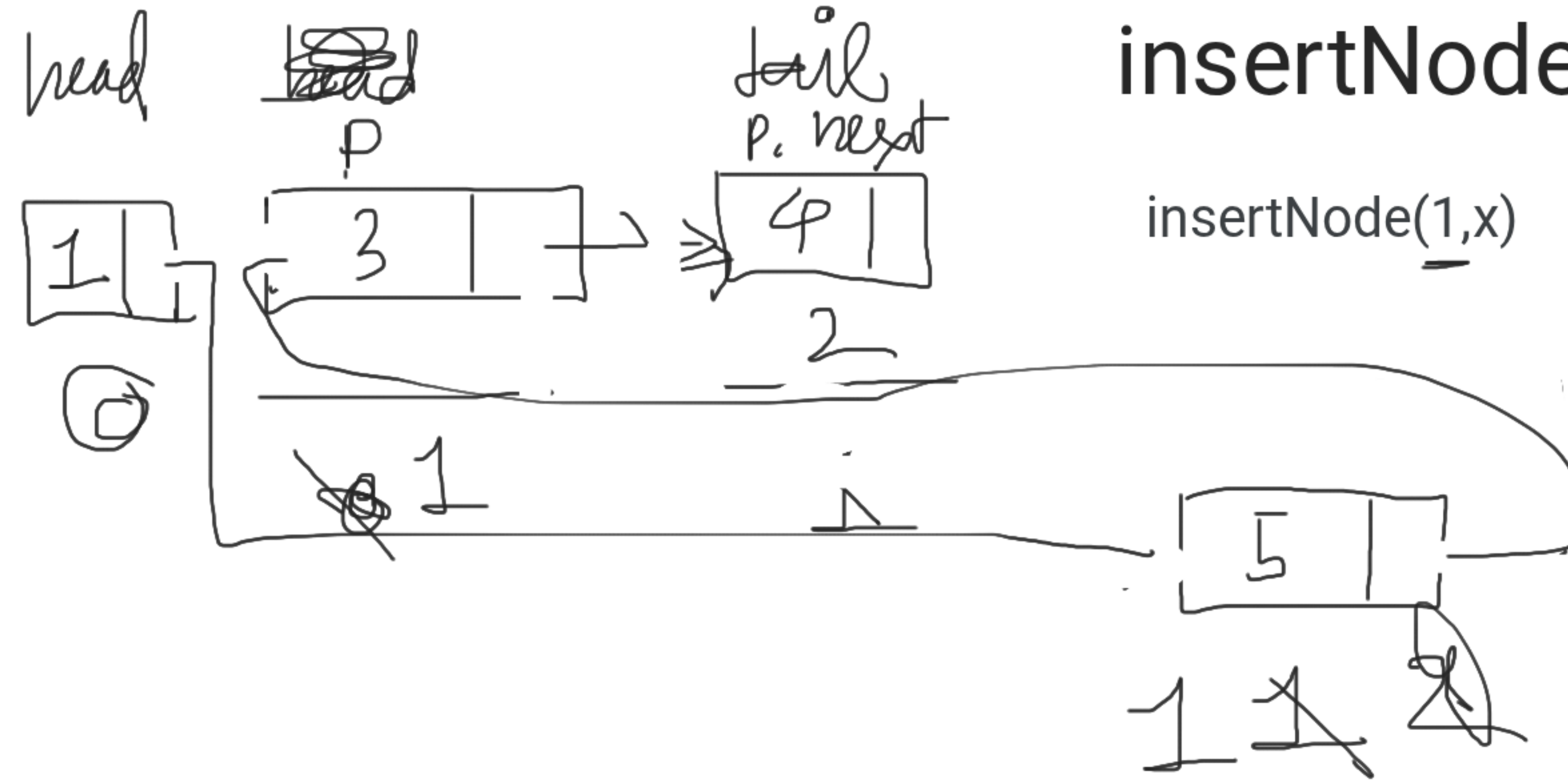
end else

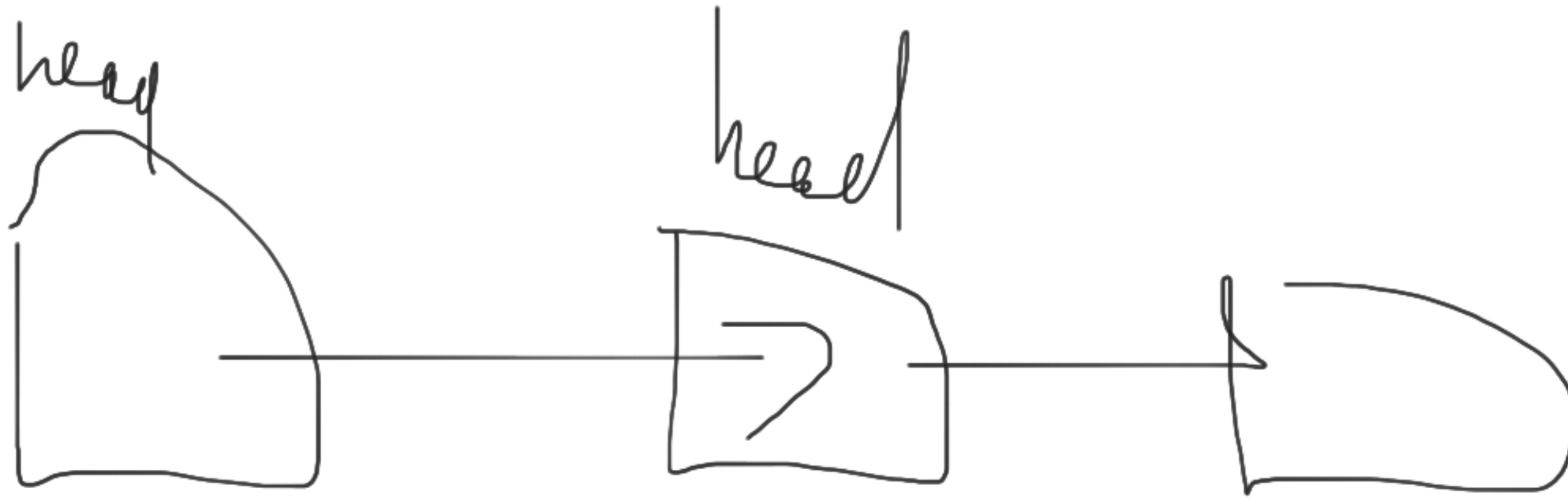
end procedure

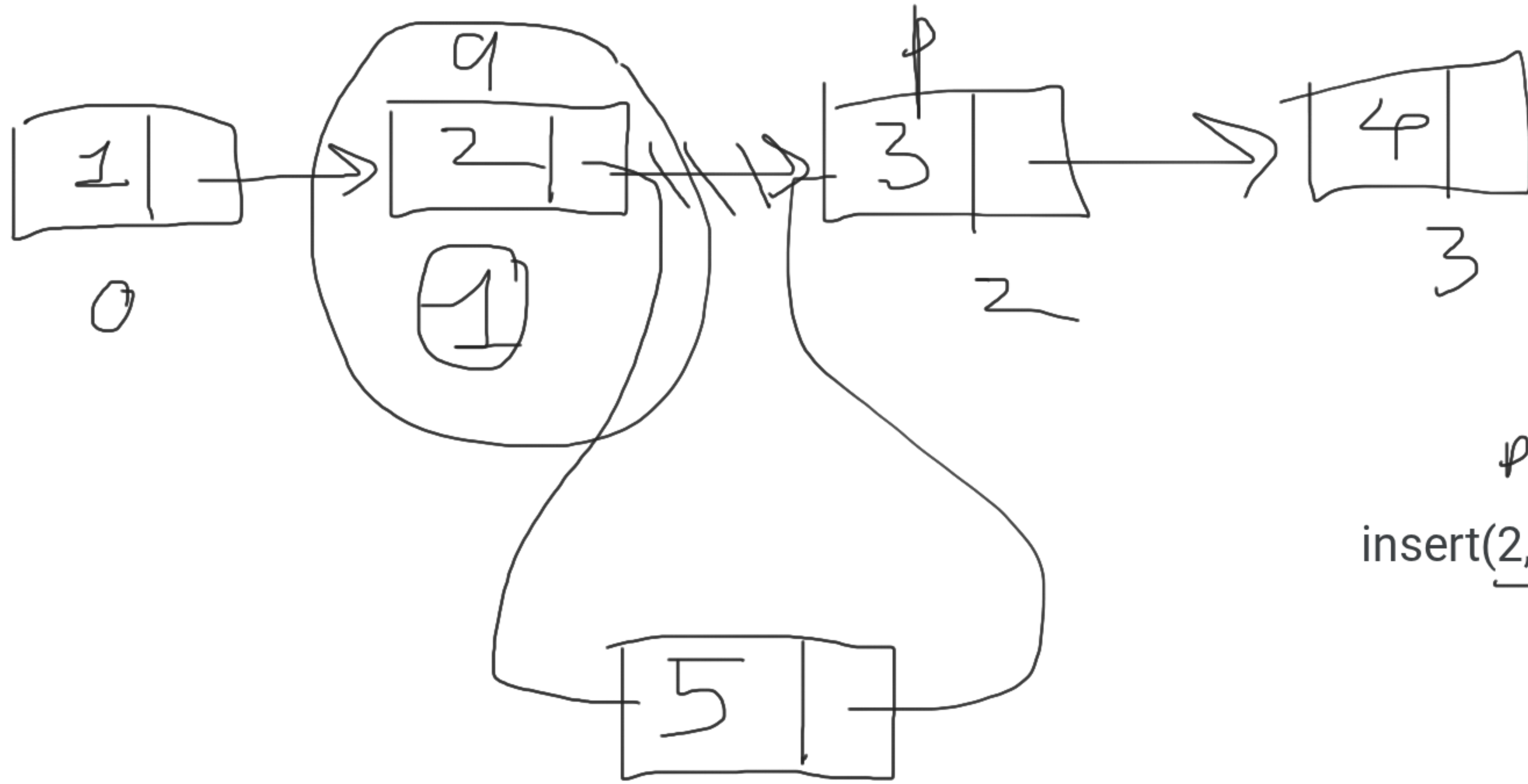
insertNode(int pos, Node x)

insertNode(1, x)

getIndex(1-1)



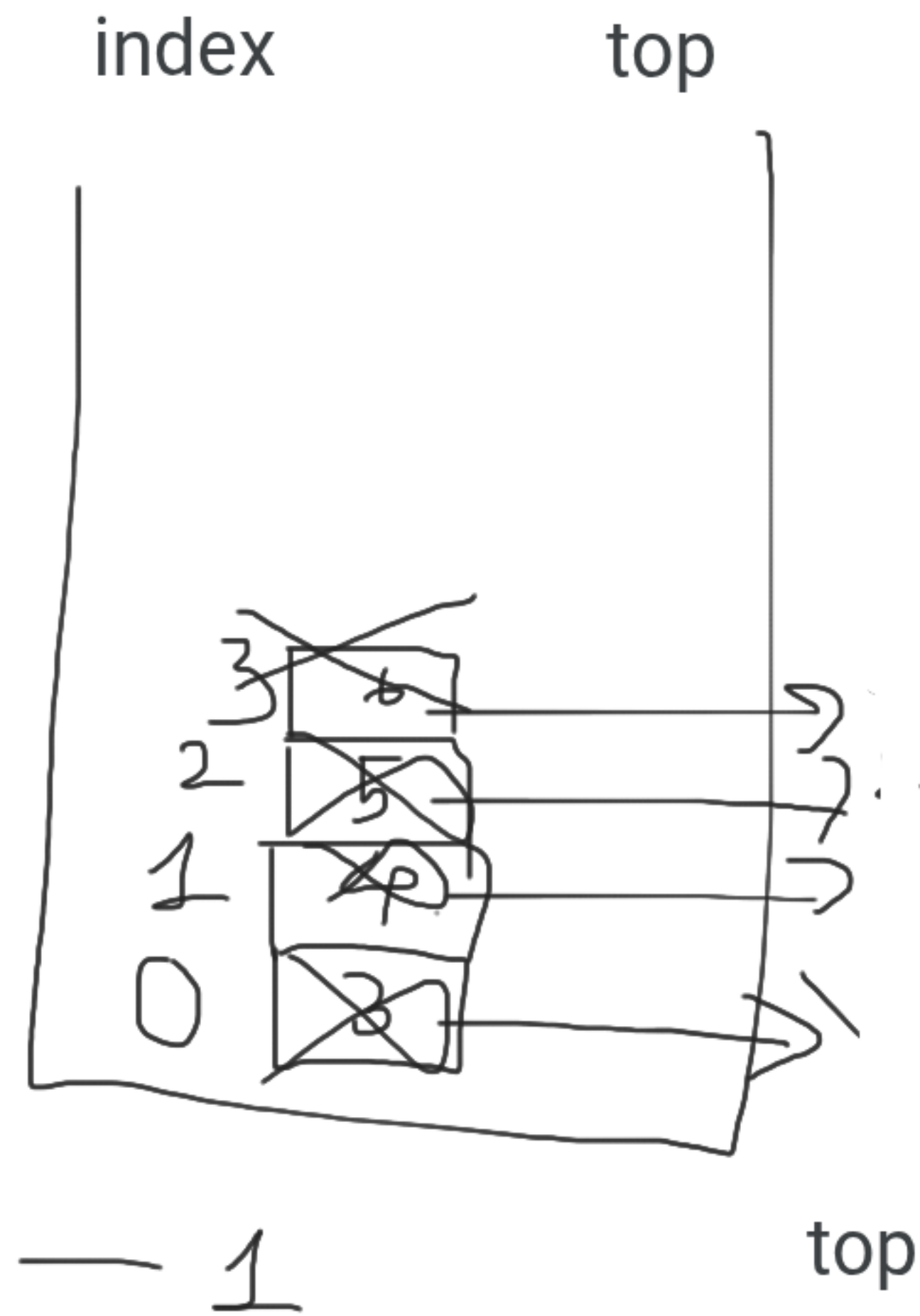




$p = q.next$
 $q.next = newNode;$
 $newNode.next = p;$

pos
 $insert(\underline{2}, Node)$

$getIndex(2 - 1)$
 $getIndex(pos-1)$



max = 4

top = -1 →

~~push(3)~~

~~push(4)~~

~~push(5)~~

~~push(6)~~

~~push(7)~~

→ stack overflow

~~pop()~~

~~pop()~~

~~pop()~~

~~pop()~~

~~pop()~~

pop()

→ stack underflow

Implement Queue

simple queue

circle queue

Array or
Linkedlist

hàng đợi ưu tiên

FIFO